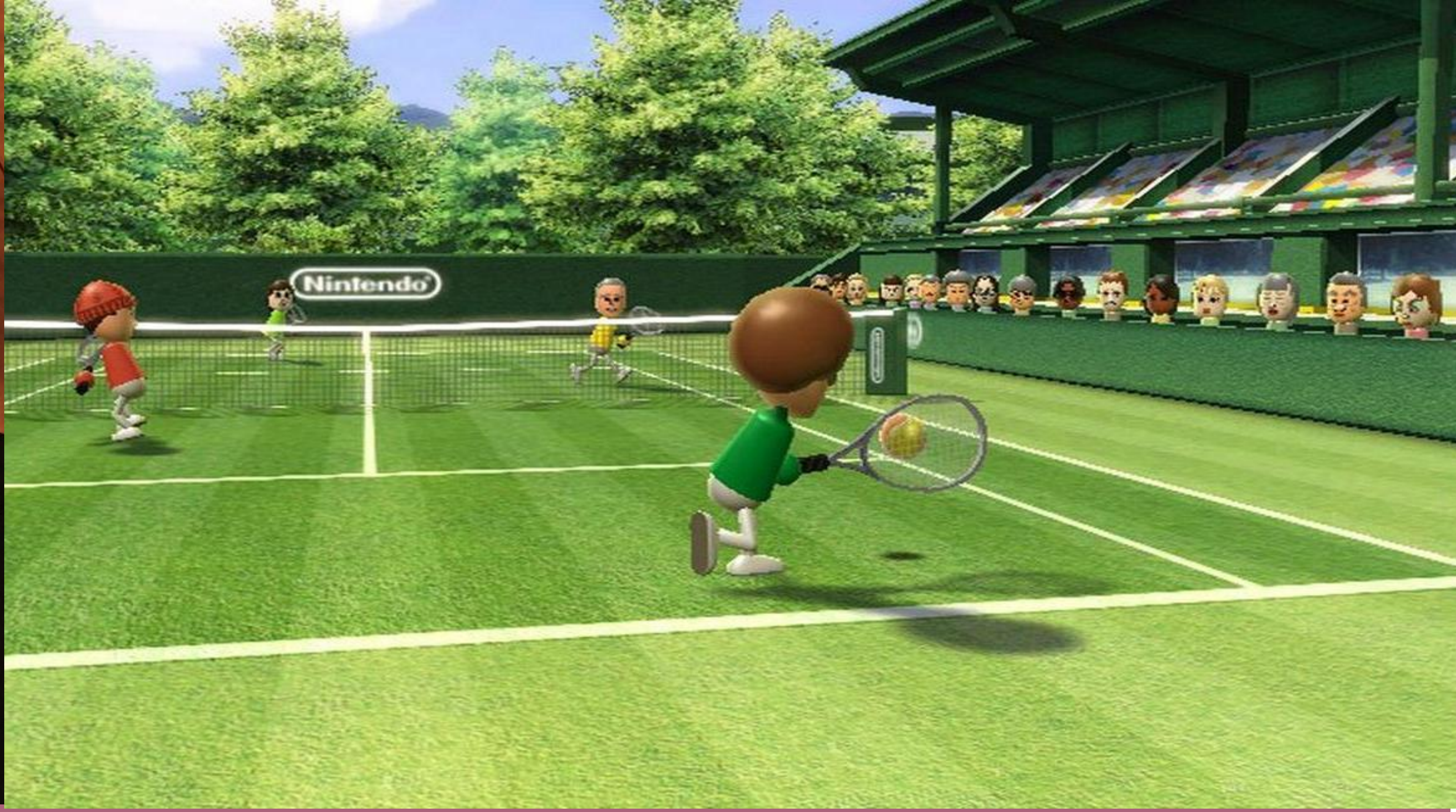


WII DID IT FIRST: WIRELESS GAME CONTROLLER

COLLEGE OF ENGINEERING
MECHANICAL ENGINEERING
VIRGINIA TECH

ME 3704 INTRODUCTION TO MECHATRONICS

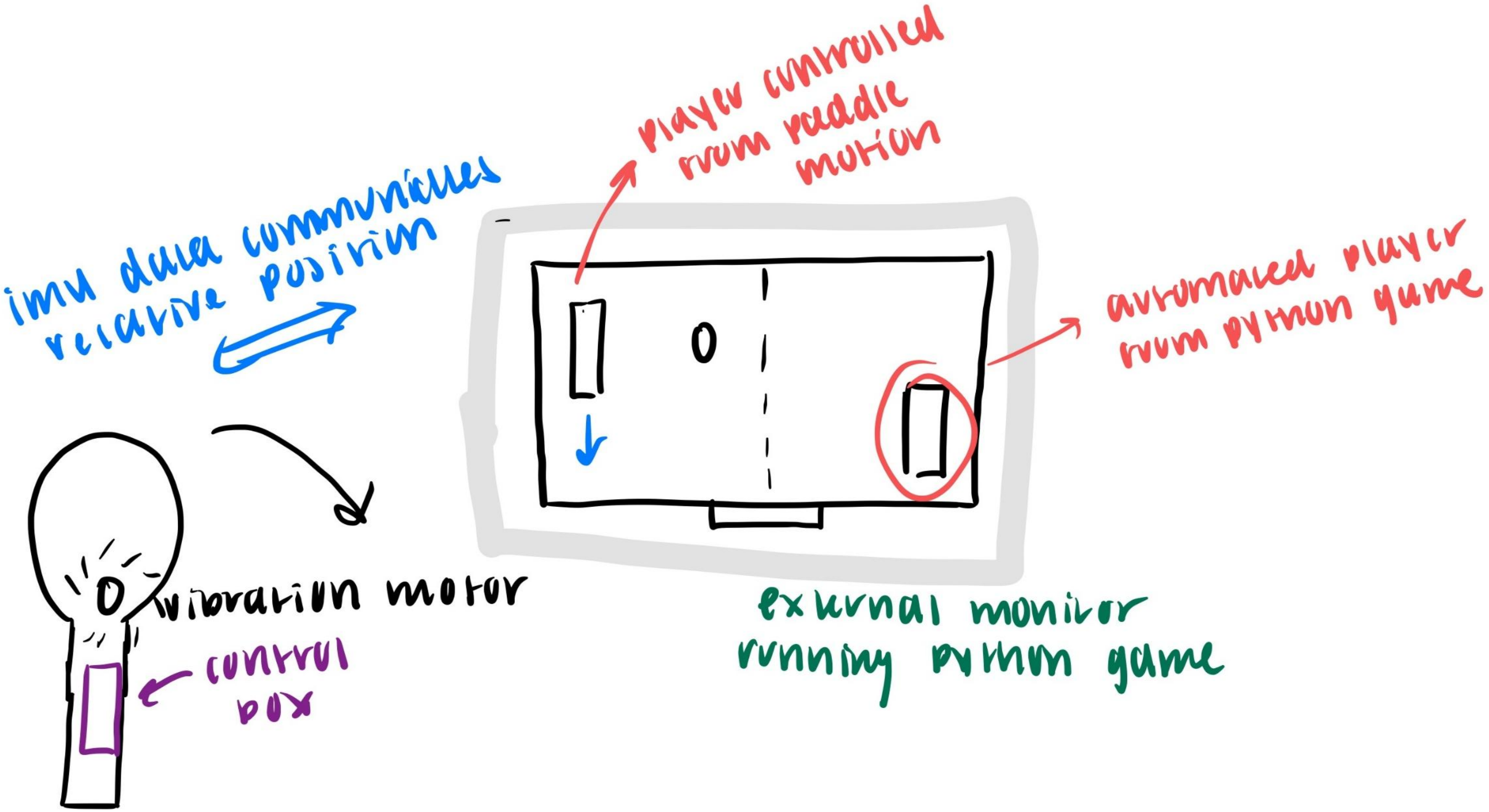
SPRING 2026 FINAL PROJECT

NATALIE MCDONALD

OBJECTIVE

- **Design and build a wireless motion–controlled game controller using an Arduino R4 and an integrated IMU**
- **Provide visual feedback to the player using Data sent from a laptop using TCP protocol**
- Main tasks
    - Control on-screen paddle position using tilt and angle from accelerometer
    - Trigger vibration motor in response to hits and update the OLED display with game data
    - Stream IMU data wirelessly to a Python tennis game
- Wanted to understand more about how sensor data fits into video game play while constructing something that I can physically use

SYSTEM OVERVIEW



- Player holds the controller, tilts it up and down to move the on-screen racket
- When they score the game sends a signal back to the Arduino built into the controller over WiFi, triggering the vibration motor

TCP AND I2C PROTOCOL

- TCP allows the Arduino and my computer to talk over Wifi, a connection is established first then both sides talk back and forth
- Arduino sends IMU data to Laptop
- Laptop sends game updates to Arduino

Originally, I tried to connect the OLED and IMU to the Arduino's default I2C pins but was having problems receiving data from the IMU – used separate buses

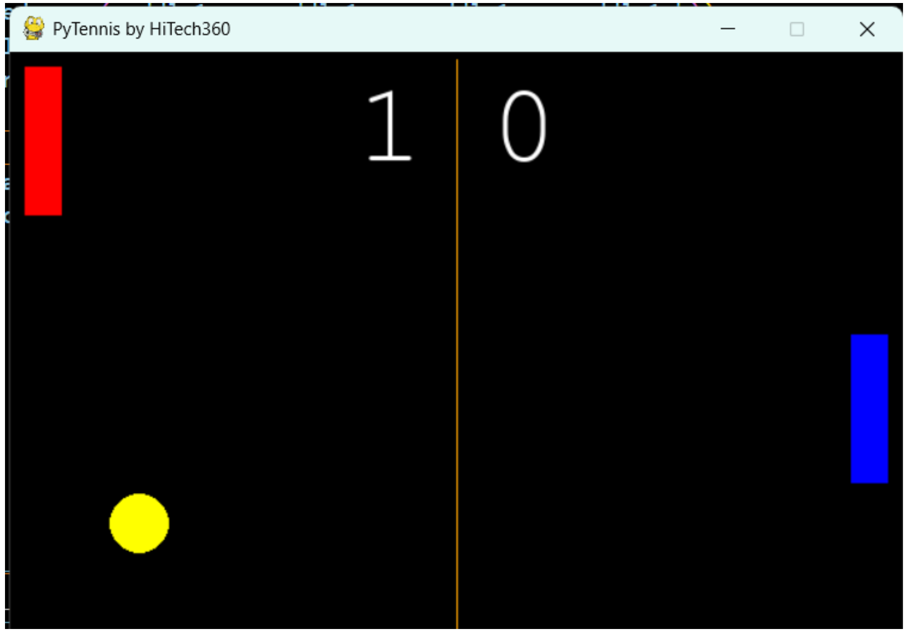
- I2C works here to allow the OLED to access data from my laptop without interfering with the IMU

DEMONSTRATION – NATALIE MCDONALD

Tilt the controller forwards / backwards to change the y acceleration and move the red player up and down on the screen

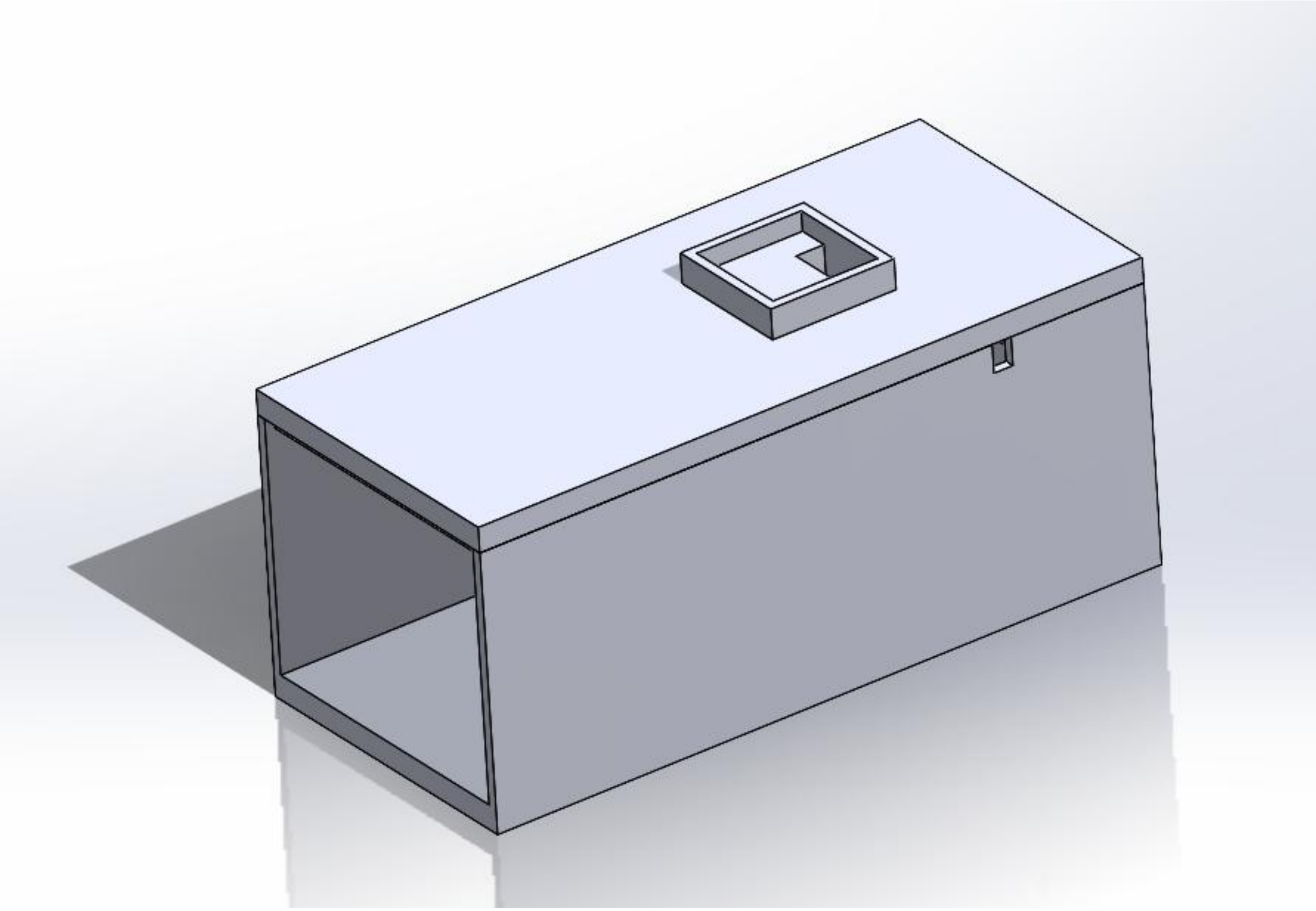
- The blue player moved automatically based on the position of the ball

When the red player hits the ball, you should feel the vibration motors buzz, and when the score updates it will be displayed on the OLED

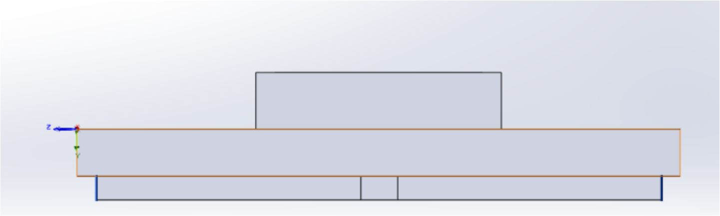


QUESTIONS?

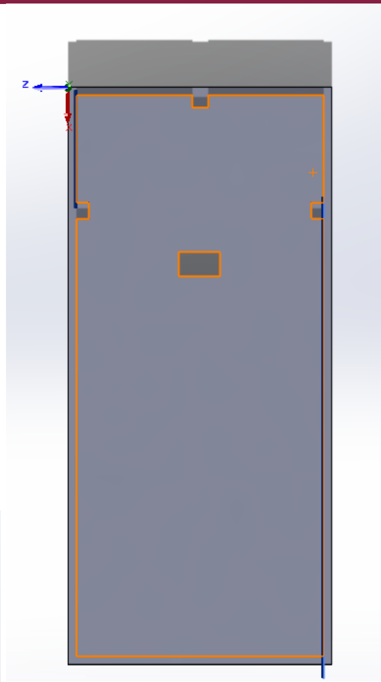
MECHANICS



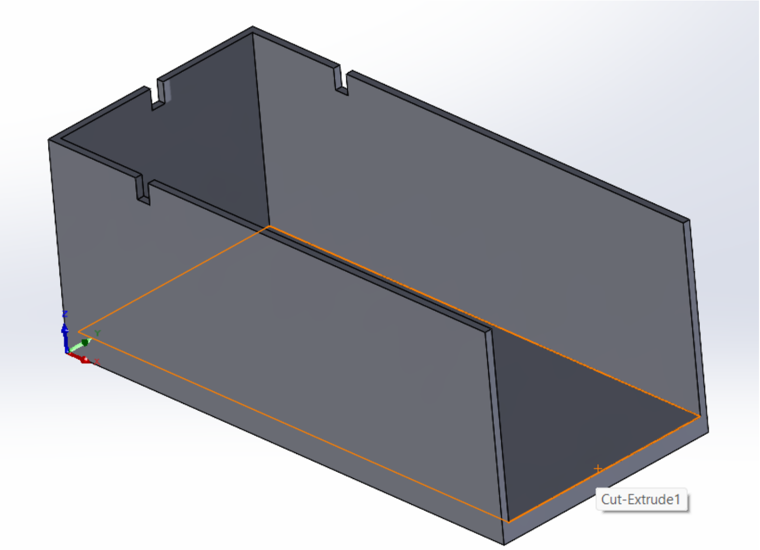
Combined housing assembly



Side view of the lid

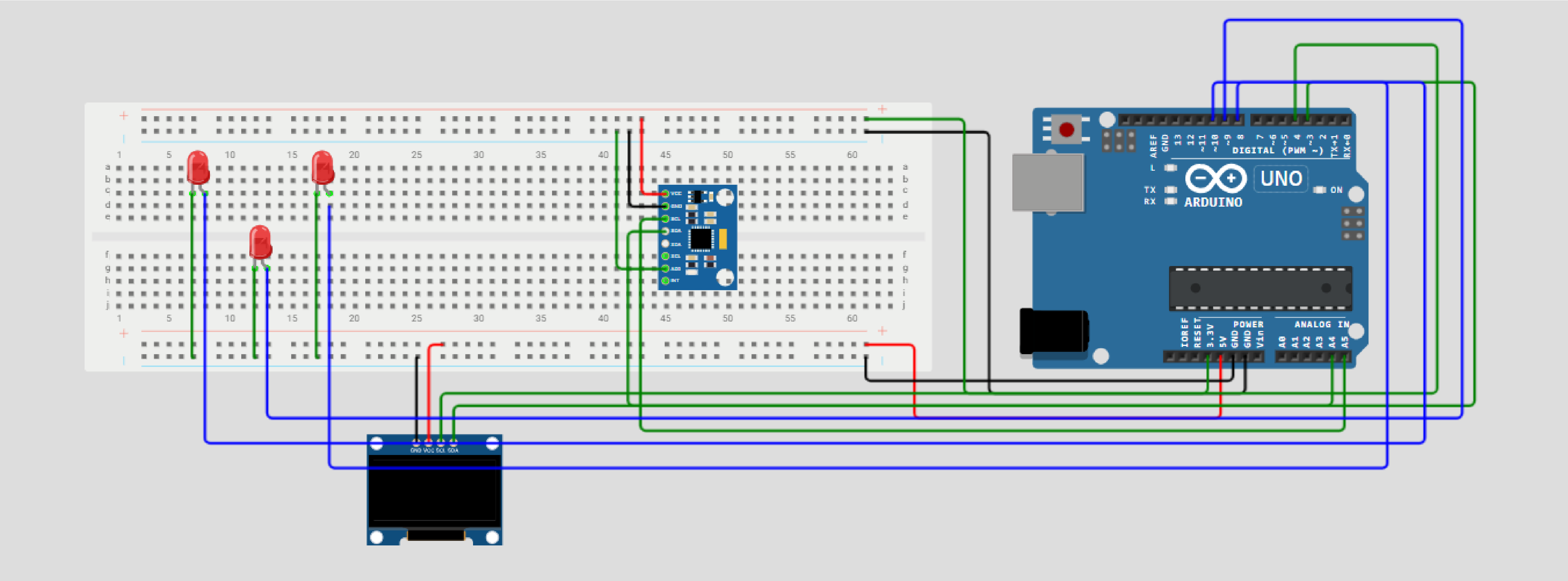


Top view of the casing



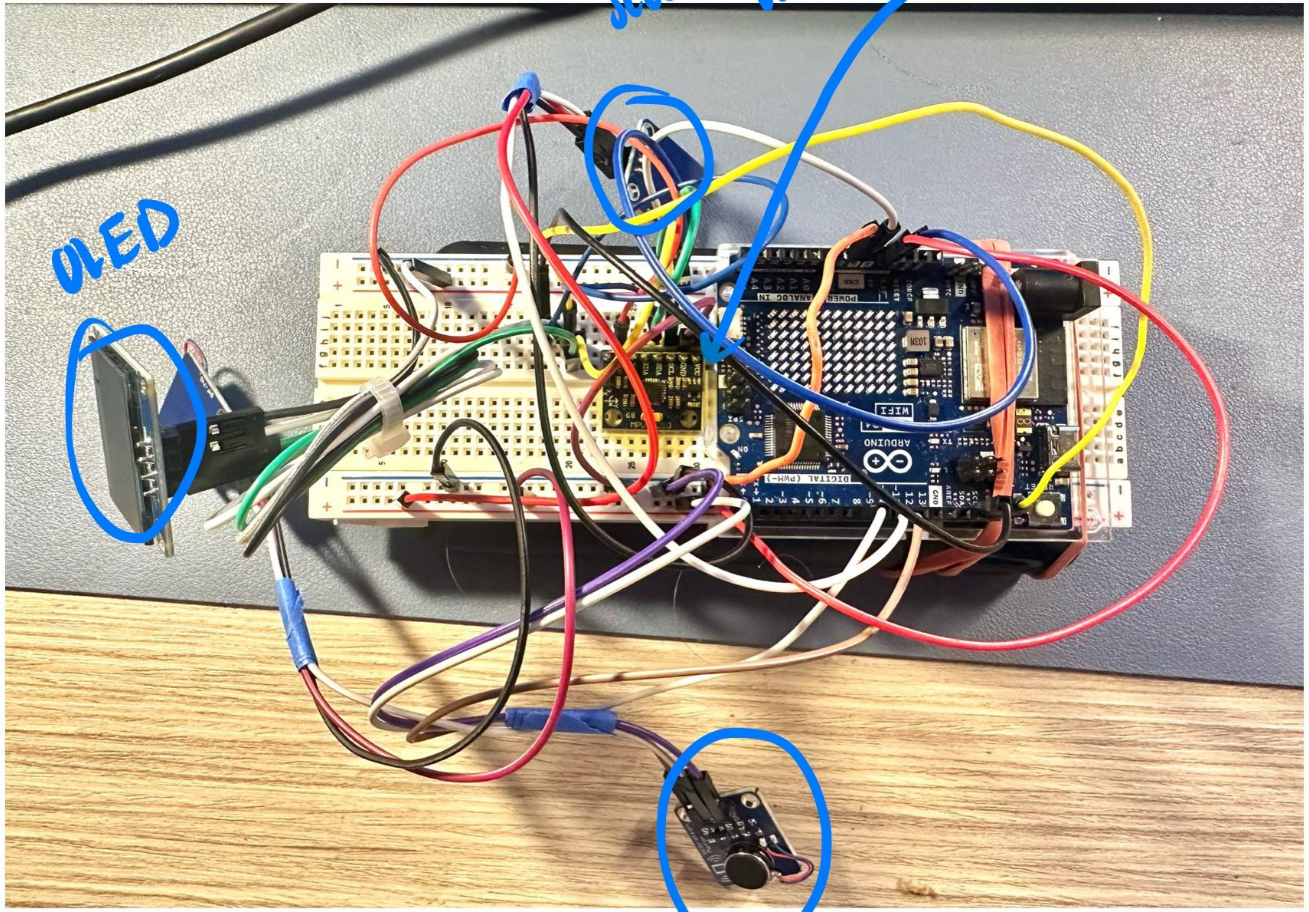
Isometric view of the bottom of the case

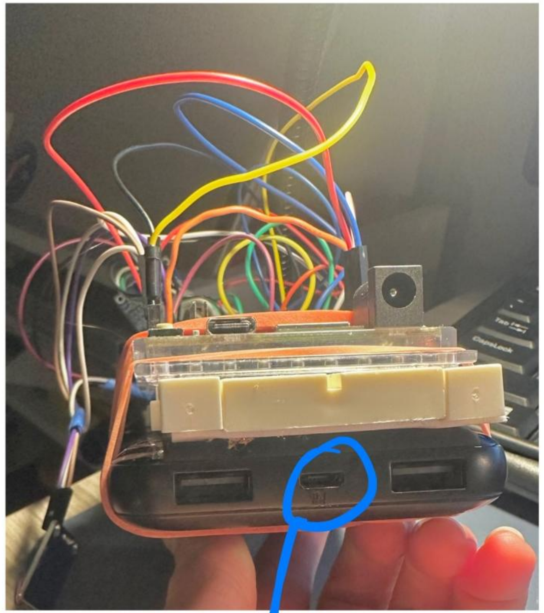
CIRCUIT



*** In this diagram the LEDs were used to represent the PWM vibration motor modules that were used in the actual prototype

CIRCUIT


second vibration motor
IMU
OLED
vibration motor


Portable 5V
Power bank

FINAL DESIGN

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BACKUP SLIDES

BACKUP SLIDES

- **Add supporting slides – as/if needed**
  - These slides are optional
  - These slides will not be graded